Experimental Neurology 233 (2012) 597–605

Contents lists available at ScienceDirect

Experimental Neurology

journal homepage: www.elsevier.com/locate/yexnr

Editorial

# Replication and reproducibility in spinal cord injury research

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ARTICLE INFO

*Article history:*
Received 23 June 2011
Accepted 28 June 2011
Available online 10 November 2011

*Keywords:*
Replication
Regeneration

ABSTRACT

This special issue of Experimental Neurology compiles a series of papers that either explicitly replicate published studies or retest phenomena reported in previous publications. The explicit replications were carried out as part of the "Facilities of Research Excellence—Spinal Cord Injury" (FORE—SCI) program launched by the National Institute of Neurological Disorders and Stroke (NINDS) in 2003. Here, we review the FORE—SCI replication experiments published prior to those in this special issue. We then discuss emerging issues regarding replication and reproducibility in spinal cord injury research, especially in terms of potential translation to clinical trials.

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## Introduction

In 2003, the National Institute of Neurological Disorders and Stroke (NINDS) launched a new program called "Facilities of Research Excellence—Spinal Cord Injury" (FORE—SCI). This program provided funding via contracts to carry out independent replication of published studies that report experimental interventions that reduce secondary injury, improve recovery, or enhance axon regeneration after SCI. Two FORE—SCI replication contracts were awarded in 2003, one to the University of California, Irvine under direction of O. Steward at the Reeve-Irvine Research Center (UCI/RIRC), and the other to the University of Miami, under direction of W. D. Dietrich at The Miami Project to Cure Paralysis (UM/TMP). The program was renewed in 2008 through a competitive solicitation, with contracts awarded again to UCI and a new contract site, The Ohio State University (OSU), under direction of P. Popovich. This special issue of Experimental Neurology compiles several papers resulting from the FORE—SCI program that explicitly replicate published studies, as well as other papers that retest phenomena reported in previous publications. Here, we discuss the organization and impact of the FORE-SCI replication program, review previously published FORE—SCI replication studies and discuss general issues regarding replication, reproducibility and potential translatability in SCI research.

### *The FORE—SCI replication program*

The FORE—SCI replication program was established by NINDS after extensive discussions among the spinal cord injury (SCI) research community of two perceived problems. First, every year for at least the preceding two decades, numerous papers appeared reporting that some intervention, drug or biologic (e.g. cell transplant), reduces injury severity or functional impairment (i.e., is neuroprotective), improves recovery of function, or enhances regenerative growth following SCI. Despite this apparent progress at the basic science level, not one of the many strategies reported as having promising effects were translated into successful therapies. Indeed, only recently was one of these interventions carried forward through preclinical development to testing in a clinical trial. Second, many reports of striking effects were never followed up either by the original authors or by others, or if they were, the results were not reported in the scientific literature. What happened? Were the results not reproducible? If there were failures to replicate, these were rarely communicated to the community at large, although rumors abound ("I tried that and it didn't work"). It is impossible to interpret or judge the validity of the rumors or whether the rumored replications were thorough and had been carried out properly.

One possibility is that the lack of follow-up indicates a lack of robustness of the findings. This may have been detected by the

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doi:10.1016/j.expneurol.2011.06.017

original laboratory or by others, and the failure to replicate is not reported in the scientific literature. Lack of follow-up could also be due to other factors. For example, obtaining funding for repeating studies or following up on previously reported phenomena might be difficult, especially with the NIH focus on innovation as a major factor in review. Another common perception is that journals will not accept studies that simply replicate previously reported findings or, more troubling, those that seek to report negative results. Finally, repeating or following up on the work of others is not the optimal way to advance one's career, especially for young investigators.

Even if a scientist chooses to carry out a replication, repeating studies can be problematic in terms of IACUC approval. IACUC applications require documentation that the proposed research does not duplicate previous work. Of course, that is exactly what a replication is. Extra justification is required to document that there is scientific value of replications that goes beyond mere duplication.

The issue of replication has been discussed in several contexts in recent years, often coupled with discussions of publication bias, transparency in reporting and questionable analyses, which impact the rigor and generalizability of preclinical studies in many disease areas (Ransohoff, 2005; Ioannidis, 2005; Scott et al., 2008). Indeed, a recent survey of SCI researchers indicates that most scientists who were queried (94%) felt that replication was critical before a novel manipulation, drug, or biological treatment could be tested in a clinical trial (Kwon et al., 2010). One important reason for this consensus is that clinical trials are vastly more expensive than basic science, and clinical trials in other fields (stroke, traumatic brain injury) have had an extremely high failure rate (STAIR, 1999; Narayan et al., 2002). There is an emerging suspicion that the basic science and/or preclinical modeling on which some of these trials were based was not sufficiently tested for reproducibility and robustness. This may have had adverse effects on the design of clinical trials, and limited not only their likelihood of success but also any informational value about whether mechanistic targets of the therapies were adequately tested in the trials. Published papers have been seen as a proverbial brick in the wall that makes up our knowledge base and upon which future scientists and clinicians can safely build. Rumors of non-reproducibility of findings challenge this view and weaken confidence in the scientific enterprise, even when they provide no evidence that can be substantiated or adequately evaluated. This is one example of publication bias (Sena et al., 2010), which will be discussed in more detail below.

The replication contracts of the FORE—SCI program were launched to determine the reproducibility and robustness of potential SCI therapeutic strategies based on publications describing promising therapies suitable for translation and, if warranted, to accelerate progress toward pre-clinical translational work and eventual clinical testing. The performance objectives for the contracts were:

1) To select and try to faithfully replicate promising, preclinical studies relating to therapies that could lead to effective treatments for human SCI.
2) To compare the efficacy of treatments in a standardized environment with a minimum of variability in surgery, animal care, outcome evaluation and cellular analyses.
3) To promptly report the methodology and results.

The desired result is that, if proven to provide reliable and robust benefit, these promising strategies would be appropriate to move to the next level of translational studies or, if appropriate, clinical testing.

*How does the FORE—SCI replication program work?*

The original FORE—SCI contracts were awarded in 2003 through competitive proposal procedures under a Broad Agency Announcement (NIH-NINDS-02-09), and the renewal of the program was solicited in 2008 under a competitive Request for Proposals (NIH-NINDS-08-02). Criteria for competing organizations under both solicitations included the ability to carry out a range of studies on SCI in multiple standard rodent models, and expertise in outcome measures that address a variety of injury and recovery mechanisms.

*Each FORE—SCI site was designated for a particular focus.* UCI/RIRC was identified as the site that would focus on repeating experiments involving chronic injuries or manipulations to promote axonal growth and regeneration in rodent models of human SCI. The UM/TMP (2003–2008) and, later, OSU centers (2008–2013) were identified as sites that would emphasize repeating experiments involving neuroprotective strategies and interventions delivered during the acute post-injury phase.

*Contracts are different from grants.* Contracts are used by NIH to acquire a defined service (e.g., the conduct of research) and set of "deliverables" for the direct benefit and use of the government, or the research community at large, in support of the Institute's mission. Activities carried out under the auspices of the contract are defined by the Statement of Work; additional studies outside the originally solicited/reviewed scope (i.e., replication) cannot be carried out using contract funds. Thus, follow-up studies that might have been conducted under a grant mechanism were not pursued unless the PIs chose to do so using non-contract funds.

*Selection of studies for replication.* Experiments are selected for replication with advice from independent Scientific Advisory Committees, one designated for each site, made up of experts from the SCI field. The PIs and Scientific Advisory Committees review the literature to identify and prioritize experiments for replication. The Committee may select only part of a published study for replication (for example a subset of the experimental data) and may also request additional experimental groups or analyses. For example, there may be additional control groups or groups that test either combination strategies that make sense within the framework of the methods of the original studies or provide a direct comparison of the original work with novel treatment groups. As noted above, the initial contract provided for few such additions, but based on experience with initial replications, this capability was expanded during the re-competition. The following factors were weighed in the selection process:

- Clinically relevant endpoints (usually means that the studies include functional analyses indicating sparing or recovery of function).
- Translatability of the treatment to the clinic.
- Degree of improvement (effect size).
- Scientific quality of the publication.
- Note is taken when all studies on efficacy of a particular intervention come from a single set of PIs.
- General strengths and weaknesses.

*Execution of the replication study*

Once the target publication is selected, the original authors are contacted to inform them of the planned replication and to clarify experimental details. It is noteworthy that the different FORE—SCI Centers approached this in somewhat different ways and engaged the original authors to varying extents. Indeed, there was a difference in involvement of the original authors across the replication studies carried out by a single FORE—SCI Center as further described below. In retrospect, the different approaches were based in part on a difference in philosophy about the most effective way to plan and execute independent replication experiments.

*Criteria for an independent replication*

When does an independent replication become a collaboration between the original and replicating authors? A theme that emerges in the replication papers in this special issue and their associated commentaries is the question of the appropriate degree of involvement

of the authors of the original papers. The goal of the contracts was to try to test exact replications to the extent possible. But the extent to which the original authors should be involved remains a point of contention.

For example, an opinion at one end of the spectrum is that the original authors should be involved in different stages of the planning and execution of the replication, even traveling to the contractor's lab to review specific methods, particularly surgeries. For some specific recommendations in this regard, see Commentary by Bunge et al. herein. But some authors may not be available or may be unwilling to cooperate. In the case of the UCI/RIRC's replication of an olfactory ensheathing cell graft study (Lu et al., 2002), the original authors were not available to travel to the U.S. For the UCI/RIRC's replication of a Nogo receptor inhibitor study (Li and Strittmatter, 2003), Dr. Li traveled to Irvine to produce the dorsal hemisection injuries, because the lesion is such a critical variable in studies of axon regeneration. At OSU, the successful replication of a glibenclamide study (Simard et al., reported herein) involved reverse site visits whereas in the replication of an LPS/pregnenolone/indomethacin combination strategy (Guth et al., reported herein) the original authors were retired and unavailable to participate directly.

Our experience indicates that involvement by the original authors certainly increases the likelihood that the replication will be carried out with increased fidelity to the original. If the original authors are willing and able to cooperate, then a decision must be made about the level of their involvement, as care must be taken to ensure that the independent experiments and analyses are not unduly influenced by the initial authors. Nevertheless, it is our view that published studies may be considered for replication whether or not there is agreement, cooperation or compliance by the original authors.

An opinion at the other end of the spectrum is that replications should be carried out in a truly independent fashion based on what is published in the Methods Section of the original paper, with minimal involvement of the original authors. The argument here is that involvement of the original authors nullifies independence. The problem is that published Methods sections are increasingly sparse and incomplete. In many cases, reference is made to previous publications for key approaches, but scrutiny reveals that the papers referenced are either incomplete themselves or actually use different methods. Thus, replicating papers published in journals with sparse methods is difficult if not impossible unless the original authors are consulted. There clearly is room here for differences of opinion.

### *Issues in replicating experimental details*

How identical do the experimental details have to be? The FORE—SCI contracts seek to replicate all important experimental details but this may be impossible, or at least extremely difficult (see below). Also, a reasonable argument can be made that the goal of testing potential therapies is better served by allowing some deviation from the exact methods used by the original authors (more on this below). The issue is whether the best form of replication also tests the scientific validity of the original findings, the robustness, and therefore the potential for translation.

*Experimental animals.* It may be considered important to use exactly the same type of subject from the same supplier (see Commentary by Bunge et al. herein). But what if the original study was conducted in Europe? Obtaining animals from a European supplier may be impossible or at least extremely costly for American, Asian or other researchers, and certainly long distance transport of animals has effects as well. Even if orders are placed for animals from a particular colony of a supplier, as in the case of the replication of a Schwann cell-cAMP study (Pearse et al., 2004; Sharp et al., herein), this does not ensure that the animals are not different from the original stocks. Suppliers claim that animals are genetically identical over time and from colony to colony, but this only means that no genetic differences are detectable by the methods used by the supplier. Genetic drift in a particular colony is certainly possible, which could lead to differences between colonies of a given supplier (see Bunge Commentary herein). Bunge et al. also note that suppliers may combine animals from different colonies to fill an order, and may not inform the investigator of this fact.

*Reagents.* What about making sure that reagents are from the same batch and lot? This may be important in principle (see Sharp et al., herein replication of an EGF receptor inhibitor study, Erschbamer et al., 2007), but the supplier may not be able to provide the reagent from the same batch and lot as used in the original study. Does this mean that the paper should not be replicated? If this argument is accepted, it would be equally fair to ask what point is there of publishing a paper using a reagent that can never be obtained again in exactly the same form? Under the contracts, we were confronted with this question and decided that it seems reasonable that if the work truly holds potential to be pursued because of the value of the target mechanism, or reported effect size, it is acceptable to consider a replication even if the identical reagent cannot be obtained.

*Surgical procedures.* What about the exact procedures for surgery (anesthetic, post-operative analgesics, etc.)? Obviously, an exact replication would use identical drug treatment protocols, especially for drugs that may alter injury parameters or post-injury responses, but this is not always possible due to local IACUC requirements. IACUCs may insist on particular anesthetics or post-operative analgesics. Does this mean that the paper should not be replicated? Again, one has to consider the likelihood that the different variables will significantly impact on the main effect that is being tested. A different anesthetic or drug could certainly alter the effect of a treatment, but if this occurs, can that treatment be considered robust? These considerations are of particular importance for any treatment that is considered a candidate for therapeutic advancement. The enormous heterogeneity of human spinal cord injuries and their causes means that treatments will have to produce robust effects in real world clinical situations if they are to have a chance of success in demonstrating efficacy in a clinical trial.

*Animal housing, and handling.* What about housing conditions and post-operative care? Details on housing are rarely provided in Methods sections. To maximize the likelihood of replication, conditions should be duplicated, but there is wide variance in animal housing conditions across institutions based on local conditions and requirements that may be well-justified. For example, at the International Collaboration on Repair Discoveries (ICORD) at the University of British Columbia, the practice is to house animals on a reverse light/dark cycle so that animals can conveniently be tested during their normal active phase. Yet, time of day of testing is rarely reported in most Methods sections. A reasonable argument can be made that if housing details are critical to achieve a therapeutic effect, then the treatment is unlikely to translate successfully to human use, as there may be wide variability in the rehabilitation and home settings for persons with SCI. Again, the issue is whether the goal is to validate the original science or assess translatability.

*Injury models.* In the FORE—SCI replications, the goal was to use the same injury models to the extent possible, including the devices used to create contusion injuries. This has largely been achieved, except for the replication of an erythropoietin study (Pinzon et al., 2008a) because the original paper (Gorio et al., 2002) used a device that was constructed locally. If it is assumed that different models may trigger somewhat different cellular and molecular responses and have different functional consequences, using the same device would seemingly increase the likelihood of a successful replication. However, the exact mechanisms by which models differ have not been well-defined, nor has the relevance of these differences to the predictive value of the model for translation. Until such mechanistic targets are identified, it seems reasonable that a viable treatment strategy should

be effective for general classes of injuries, for example for "moderate contusion injuries", rather than being specific to different devices. This is an issue for discussion by the SCI community at large and there is room for differences in opinion. In any case, the decision stipulated in FORE—SCI, to use the same injury models as the original papers, has had consequences in terms of the need for preparation and practice and variability in outcome as discussed in the different replication papers.

*Reducing or managing variability.* How much preparation should there be to try to reproduce exactly the same injuries and outcomes, with the same variance across animals as originally reported? Minimizing variance may be important to increase the likelihood of detecting a statistically significant effect, so most investigators seek to minimize experimental variability in their studies. This does not accurately model the heterogeneity of the clinical situation, however. Failure to reproduce the reported variability must be considered when evaluating a replication study in which there is no statistically significant effect. Indeed, if the replication reveals a similar trend as the original study, this can be viewed as support for the original biological finding. Nevertheless, if an effect can be seen only when variability is tightly controlled in a laboratory setting, one must question whether the intervention could be the basis for successful translation.

Even though the replication centers were selected specifically as sites with broad expertise in SCI models and facilities for implementing a wide variety of study designs and outcome measures, it has been necessary to spend substantial effort to match parameters of injury severity with the anatomical or behavioral outcome measures reported in the original studies. It has also been necessary to incorporate additional models at each center (e.g., the clip compression injury) and attempt to achieve the level of reliability required to perform a successful replication. Matching injury parameters and outcomes between laboratories is an issue, even when the same injury device or type of injury is used. The issues of practice with an unfamiliar injury device and variability in outcomes arose in the UCI/RIRC replication (herein) of the Pearse et al., (2004) study and the OSU replication of Simard et al. (2007) (see Popovich et al., herein). At UCI/RIRC, the Infinite Horizons (IH) device (Scheff et al., 2003) is the instrument of choice for creating moderate contusion injuries; at OSU, this is also the primary injury model. The IH device produces primary tissue damage

**Table 1**
Summary of FORE-SCI replication studies.

| Original article | Original finding | Result of replication |
|---|---|---|
| Lu et al., 2002 | Delayed transplant of olfactory lamina propria (OLP) improved hindlimb motor function after complete transections in rats. | No replication. No significant improvement in hindlimb function in rats that received OLP transplants. |
| Li and Strittmatter, 2003 | Intraspinal delivery of NEP1–40 improves hindlimb motor function and enhances CST sprouting after thoracic dorsal hemisection in mice. | Partial replication. Enhancement of locomotor function in one of two duplicate studies; no difference in CST axon growth. |
| Pearse et al., 2004 | Combined treatment with Schwann cell transplants, Rolipram and intraspinal injection of dbCAMP improves locomotor recovery after thoracic contusion in rats. | Mixed results. Rats that received Schwann cells only improved, but the combinatorial treatment was not significantly better than single treatments. |
| Erschbamer et al., 2007 | Intraspinal delivery of an EGF receptor antagonist (PD168393) enhances recovery of hindlimb motor and bladder function after thoracic contusion in rats. | No replication. Treated group was significantly more impaired and lesion size was larger. |
| Bradbury et al., 2002 | Intraspinal delivery of ChASE allows regeneration of CST axons following cervical dorsal crush injuries in rats. | Inconclusive because lesions spared CST in some rats. Not repeated because of other studies supporting original study. |
| Gorio et al., 2002 | Delivery of recombinant Human Erythropoietin (EPO) reduced injury severity and improved locomotor recovery after thoracic contusion and compression in rats. | No replication. No significant effect of treatment. |
| Lee et al., 2004 | Minocycline treatment reduces cell death and improves hindlimb motor function after contusion injury. | No replication. No significant effect of treatment. |
| Gris et al., 2004 | Treatment with a monoclonal antibody to the CD11d integrin subunit reduced infiltration of neutrophils, improved neurological outcomes, reduced neuropathic pain and histopathological damage following clip compression injury in rats. | Partial replication. There was a trend for greater recovery and reduced tissue damage, but differences were not statistically significant. |
| Wang et al., 2004, Peng et al., 2009 | Intraspinal delivery of P2X7 receptor blockers or systemic administration of the P2X7 receptor antagonist Brilliant Blue G improved hindlimb locomotor function and reduced injury severity after thoracic contusion in rats. | No replication. Treated groups did not differ significantly from controls. |
| Simard et al., 2007 | Delivery of glibenclamide, which targets (SUR1)- regulated cation channels, attenuates secondary intraspinal hemorrhage and neurodegeneration following cervical hemi-contusion injury in rats. | Successfully replicated after discovering that the effect depended on the exact mechanism of injury. |
| Guth et al., 1994 | Acute treatment with a combination of pregnenolone, LPS, and indomethacin enhanced hindlimb locomotor function and reduced lesion size after thoracic crush injury in rats. | Replicated but with less robust effects. Differences in outcomes assessment, drug composition and injury model may have degraded robustness of effect. |
| Benowitz et al., 1999 | Intracranial delivery of inosine triggers trans-midline sprouting of CST axons after unilateral injury of the medullary pyramid in rats. | No replication. No evidence of trans-midline growth of CST axons in inosine-treated rats. |

that is somewhat different from that of a conventional weight drop device, which might affect outcome and success in replication so it was deemed important to use that device for the replication of Pearse et al., which used the "NYU" device (Constantini and Young, 1994), though there was greater variability in the lesions and outcomes in the UCI/RIRC replication as discussed in the paper (Sharp et al., herein; see also Commentary by Bunge et al. herein).

The original contract did not anticipate the need for "practice" studies, however, and the time required to become proficient with unfamiliar injury models can be considerable, especially because one issue is variability in lesion size, which requires histology and detailed quantitative analysis. Also, such preparatory experiments may be problematic with IACUCs; extensive practice flies in the face of the 3R's for animal experimentation (Reduction, Replacement and Refinement). The question remains, is there a reasonable criterion for acceptable variance in a particular animal model, in either original or replication studies, regardless of how many animals must be prepared? Choices must be made, and there is room for differences of opinion.

*Previous publications from FORE—SCI replications.* Several replication studies have been published prior to this special issue, and are summarized here along with the other FORE-SCI replication studies in this special issue (see Table 1).

*Previous publications from the Reeve-Irvine Research Center*

1) Replication of a study reporting that transplantation of olfactory lamina propria (OLP) containing olfactory ensheathing cells one month after a complete transection of the spinal cord at the thoracic level resulted in significant improvement in hindlimb motor function and induced regeneration of descending axons (Lu et al., 2002). Our results (Steward et al., 2006) did not confirm the findings of the original publication. Assessment of hindlimb locomotor function with the BBB scale (Basso et al., 1995) revealed no significant recovery after OLP transplantation and no significant differences between OLP vs. respiratory lamina propria (RLP) transplant groups. We also did not find evidence for enhanced axon regeneration. Our interpretation was that although OLP transplants may stimulate regeneration under some circumstances, the effect is not robust enough to reliably overcome the hostile setting created by a complete transection.
2) Replication of a study reporting that delayed subcutaneous treatment with the Nogo-receptor (NgR) antagonist peptide NEP1–40 (Nogo extracellular protein, residues 1–40) enhanced locomotor recovery after thoracic SCI, induced extensive growth of corticospinal axons, and increased sprouting of serotonergic fibers (Li and Strittmatter, 2003). Our results were mixed (Steward et al., 2008a). The functional analyses revealed that mice treated with NEP1–40 had higher BBB and BMS (Basso et al., 2006) scores than vehicle-treated controls at the early post-injury testing intervals, but scores converged at later intervals. In this regard, our results partially confirmed the original report of enhanced locomotor function. Evidence of regenerative growth of corticospinal tract (CST) axons past the lesion via the ventral column was seen in all groups.

*Previous publications from The Miami Project to Cure Paralysis*

1) Replication of a study reporting that recombinant human Erythropoietin (EPO) resulted in impressive recovery, locomotive behavior and histopathological changes in a rat model of thoracic spinal cord contusion and compression (Gorio et al., 2002). Our results (Pinzon et al., 2008a) did not confirm the findings of the original manuscript. EPO treatment following either thoracic contusion or more severe compression injury failed to significantly improve hindlimb locomotor function (BBB scale) or reduce necrotic changes in the injured spinal cord. Our interpretation was that the use of recombinant human EPO needs to be further examined in clinically relevant animal models including dose response and therapeutic window studies.
2) Replication of a study reporting that minocycline treatment significantly improved locomotive behavior and reduced histopathological injury in a model of contusive SCI in rats (Lee et al., 2003). Our results (Pinzon et al., 2008b) did not confirm the findings of the original manuscript. There were no significant differences in locomotion scores obtained with BBB testing or significant differences in spared tissue or cavity area with minocycline treatment. Our interpretation was that although minocycline may be protective in some models of SCI, the particular dose that was used in this study failed to produce significant effects on clinically relevant outcome measures. Again, more preclinical data with dose response and clinical therapeutic window investigations are needed before moving this treatment into the clinic.

*General conclusions: a surprising preponderance of failures to replicate*

In the studies published to date, there is a surprising preponderance of failures to replicate. The same applies to the replication studies reported in this special issue. Why has there been a failure to replicate and what does it mean to the field? There is no simple answer.

Possible reasons for a failure to replicate include: 1) The original findings are incorrect, for example due to type I statistical error; 2) The replication was not exact, and differences in experimental details led to different results that may raise important mechanistic questions; 3) Effects were not significant because of greater variability in the replication (this would explain failure to detect an effect but not observations of different effects); 4) Those carrying out the replications were not sufficiently experienced in the injury models or methods for assessment; 5) There were differences in experimental animals (sources, housing, genetic drift); 6) There were differences in the lesions; 7) There were differences in animal care and use protocols (housing, handling, lighting, season, anesthetics, post-injury analgesics); 8) There were differences in reagents (especially problematic for cellular therapies); 9) There was inadvertent and unrecognized bias; 10) The effect is not robust (encompasses some or all of the above); 11) Finally, the possibility of scientific misconduct cannot be discounted.

Many of these possibilities are discussed in more detail in the papers describing the respective replications. It remains for the field to decide whether the scientific, in particular the preclinical, enterprise can be satisfied with such explanations. These issues are not unique to the field of SCI research, and though a number of fields have drafted guidelines for preclinical studies conducted prior to translation, many of the technical issues noted above, and in the papers that follow, deserve further discussion by the SCI research community.

An issue that goes beyond experimental techniques should also be considered: that of inadvertent and unrecognized bias.

> *"Bias is unintentional and unconscious. It is defined broadly as the systematic erroneous association of some characteristic with a group in a way that distorts a comparison with another group… The process of addressing bias involves making everything equal during the design, conduct and interpretation of a study, and reporting those steps in an explicit and transparent way"* (Ransohoff, 2010).

There are several uses of the term "bias". One can be overtly biased in attitude or approach; although important, this is not the form of bias we address here. Experimental techniques may be biased in ways that are not immediately recognizable. For example, there are several ways that simple counts of elements in histological preparations can be biased. Counts of any element (cells for example) in a histological section depend on where the counts are made and the size of the element being counted. It is for this reason that "unbiased sampling" techniques have been developed. Less often discussed are details of procedure that may create a biased outcome.

In SCI research, there are many ways that procedural bias can creep in unrecognized. For example, there may be a tendency to be more careful with the "experimental" group during the surgery. There may be non-random order of surgery or treatment so that important or difficult procedures are done first. Fatigue of the surgeon may affect performance when doing multiple procedures over the course of a day, which matters if different groups are done sequentially rather than randomizing animal order. The same consideration applies to post-operative testing. Blinding is important, but this can be difficult for long-term experiments. Investigators working with animals over a long time period come to know the individual animals. Interventions that require extended treatment should be controlled so that the experimenters remain blind, which can be challenging and sometimes impossible (for example, rats treated with the P2X7 receptor antagonist Brilliant Blue G are blue for a few days after treatment). Approaches used to reduce/control for unintended bias should be reported in Methods sections, but in the end, it may be impossible to eliminate all sources of unintended bias. The issue of unintended and unrecognized bias is not unique to SCI research (Ransohoff, 2005).

*Lessons learned and issues arising*

Our experience to date in the FORE—SCI replication program has provided useful lessons and revealed issues that are important for the field to consider. 1) Most Methods sections in modern journals are incomplete and some are misleading; 2) Exact replications are impossible; 3) Communication with the original authors is important, but how much is too much? 4) Pilot experiments are critical, and duplicate studies are helpful; 5) Some techniques are highly specific to the operator and are essentially impossible to replicate across sites (which is important to note if multicenter clinical trials are envisioned); 6) Publishing negative results is doable, important, and valued by the field; 7) Replications often lead to new discoveries, adding value to the undertaking (more on this below). In what follows, we discuss each of these issues and the lessons learned.

*1) Absence of detail in Methods sections.* Economic issues for journals have led to pressure to condense Methods sections. Indeed, at least one high impact journal now only publishes methods as supplemental material. This means that modern Methods sections almost never provide sufficient detail to allow an exact replication. This means that details regarding the original experiment must be obtained through communication with the original authors. It is important to recognize, however, that it is the choice of the original authors whether or not to provide critical details. This may be especially problematic in the case of publications from non-academic entities, though no FORE—SCI replications to date have addressed such studies.

Methodological descriptions may also be inadvertently misleading. For example, experimental groups in SCI studies are often compiled from a collection of many independent experiments done at different times; small cohorts of animals are prepared and then groups are compiled by combining cohorts. Methods sections that do not make this clear imply that all groups were tested together. Should the exact order of original testing be replicated and, if so, how? This raises the question as to whether it is good practice to compose groups from cohorts of animals prepared at different times. This may be the only choice for small labs or studies addressing complex combination therapies where limited availability of cell populations or certain reagents may limit flexibility. Also, one can argue that trying to operate on many animals during a short time period (over the course of a day for example) increases the likelihood of operator error. Again, there is room for a difference of opinion. At a minimum, we recommend that journals and reviewers require that details on batching of experimental groups be made clear in Methods sections.

*2) Exact replications are impossible.* Or, "if it's not one thing, it's another". Try as hard as you may, there will always be differences between studies — lesions, post-injury treatment, bladder expression protocol, timing of locomotor testing, time of day of testing, housing, gentleness of handling, phase of the moon! It is just not possible to duplicate every variable, nor is it possible to predict a priori which variables may later be seen as critical. In fact, this is also a key issue in the conduct of clinical trials; failure of a large trial may lead to questions about whether the choice of dose, timing, or other parameters negatively impacted the results. Because of this, emphasis is now being placed on the value of "adaptive trial designs" (Muller and Schafer, 2001; Biwas et al., 2009). While these require sophisticated statistical modeling to perform well, there may be lessons for preclinical and translational studies to consider.

*3) Communication with the original authors is ideal, but how much is too much?* Issues surrounding this question were discussed above. There are factors that limit opportunities for interaction, and there is room for a difference of opinion on how much interaction is desirable for a strict replication study. Ultimately, the question is important outside the limited context of the FORE—SCI replications. With current Methods sections being too short, and the countless variables described above working against the replicator, it is increasingly difficult for anyone trying to build upon the scientific literature by confirming a reported therapeutic effect. On the other hand, if the effect is robust enough, it should be seen even if there are minor differences in experimental conditions. Discovering that a therapeutic approach has limited generalizability, however, is an important finding that calls for further explanation if the approach is to be pursued to translation.

*4) Pilot experiments are critical.* It is important to carry out a sufficient number of pilot studies to test drive and establish experimental protocols. Experiments involving cell transplants are extremely complicated undertakings. Even for labs with considerable expertise in SCI research, such complicated experiments benefit from preliminary studies. For example, in the UCI/RIRC's replication of Lu et al., 2002, there were difficulties creating complete transections using the surgical procedures described in their paper (simply passing a blade through the spinal cord). Indeed, it is well-known in the field that this approach often spares the ventral most portion of the spinal cord, and this was the case in some of the animals. Modifying surgical procedures in a subsequent experiment resolved the problem, but it would have been more efficient to carry out pilot studies to make sure that the surgical procedures actually produced complete lesions as the study required. A problem of animal attrition due to autophagia was also eliminated in the duplicate experiment by modifying the post-injury treatment protocol, but this too could have been alleviated through pilot studies.

While there was no provision in the FORE—SCI contract Statement of Work for pilot studies, in retrospect it seems some pilot studies could have saved time or expense. The replication situation is not unique; issues of incomplete (or incompletely documented) lesions and inter-rater reliability in use of outcome measures also affect the reliability of original studies, though they are rarely addressed directly. In fact, specific technical details considered critical to success should be clearly described as such in original Methods sections, though the replication experience has shown us that this is rarely the case. NINDS has attempted to address the question of expertise in SCI laboratories through other components of the FORE—SCI program: a hands-on training course at OSU (2003–2013), and an advanced training opportunity at the UM/TMP (2003–2008). This remains a concern, however, that the field as a whole must address.

*5) Some techniques are more art than defined protocol, and are difficult to replicate across sites.* The move over the past decade to standardize injury devices has helped this problem. Nevertheless, some studies still use procedures that are difficult to standardize. The Miami Project used the NYU impactor well-characterized by the MASCIS group (Basso et al., 1996) for the replication of Gorio et al. (2002) rather than a device developed at The University of Trieste as was used in the

original studies. Other examples include partial crush injuries where only a part of the total circumference of the spinal cord is crushed. The UCI/RIRC completed a replication of the report on chondroitinase (Bradbury et al., 2002) at the end of the first contract period, but the results were complicated by the fact that our partial dorsal crush injuries spared some CST axons in many animals in the replication experiment, which complicates the interpretation. Clip compression injuries, developed and used more extensively in Canadian laboratories (Fehlings and Tator, 1995), have also been difficult for several other groups to incorporate and standardize. Even so-called standardized contusion devices, including those that are commercially available, can produce variable injuries. The surgical preparation (extent of the laminectomy) and method of stabilization of the spinal column (clamping) are critical variables. Also, the standardized contusion devices contain electronics and moving parts that need to be replaced periodically, which may affect performance as they sustain wear and tear.

*6) Publishing negative results is doable, important, and valued by the field.* When the FORE—SCI replication project was launched, there was considerable discussion about whether papers describing replications would be accepted for publication and valued by the field, especially if the results were negative. Contrary to this concern, our experience has been that the studies have been well-received by reviewers and journal editors and are considered to be important for the field.

*7) Replications can lead to new discoveries.* The replication studies by the UCI/RIRC led to two unexpected discoveries. One was a BDA labeling artifact that accounts for a large part of the discrepancy between three published studies of regenerative capacity in Nogo knockout mice (Steward et al., 2007). The other was a previously unrecognized form of CST regeneration that occurs in control mice (Steward et al., 2008b). The replication of Simard et al. by OSU reported herein led to the discovery that an agent that reduces hemorrhagic necrosis is most effective when the primary hemorrhage caused by SCI is moderate. This is important information to guide patient selection for any clinical trial designed to test this therapeutic. Indeed, this information alone could make the difference between a failed vs. a successful trial. For treatments that are applicable in only a limited subpopulation of patients, not taking heterogeneous responses into account in the design or analysis will preordain failure for a clinical trial. Assessing whether a therapeutic strategy is replicable and generalizable, therefore, can help identify in advance the subset of patients who are most likely to benefit, minimize unnecessary risk to trial participants and increase the likelihood of trial success.

*Questions about replicability are being recognized in many fields.* There is a growing interest in the issue of replicability and robustness of effect in epilepsy, stroke, traumatic brain injury (TBI), amyotrophic lateral sclerosis (ALS) and other fields. This interest is driven in part by a large number of failures of early stage clinical trials that were based on promising findings in animal models. A recent example involves the report that amyloid beta may act by binding to cellular prion (Lauren et al., 2009). Three follow-up studies failed to confirm the key findings, however, including one that replicated the measurements used in the original paper. Quotes in a Nature News and Views article (Ledford, 2011) regarding the discrepancy mirror the issues discussed here: “There's just a history of this in the field. There are these spectacular results and they have a half-life of 6 months to a couple of years” (Roberto Malinow). “May be due to differences in animal models, memory tests, and methods of preparing and administering amyloid beta” (Steve Strittmatter). “If it depends so much on the original experimental conditions, then the findings become no longer necessarily relevant” (Gianluigi Forloni).

In the ALS field, the issue has come to center stage in terms of studies involving SOD knockout mice (Scott et al., 2008). The study re-assessed 70 drugs that had been reported to increase lifespan in an animal model of ALS (SOD1-G93A knockout mice). Drugs that were re-tested included minocycline, creatine, celecoxib, sodium phenylbutyrate, ceftriaxone, WHI-P131, thalidomide, and riluzole. Two hundred twenty one separate studies were carried out over a five year period that involved over 18,000 mice. For some analyses, the re-assessment involved a larger number of mice than all of the previous studies of that drug combined. Rigorous and appropriate statistical methods were used. A previously described gender effect was confirmed, but there was no statistically significant positive (or negative) effect for any of the 70 compounds tested. The remarkable conclusion was that previously published positive effects were measurements of noise in the distribution of survival means as opposed to an actual drug effect (for further discussion, see Sharp et al. herein, replication of Pearse et al., 2004). Importantly, some of the agents that were retested are already in clinical trials.

*Is independent replication required to go forward with an FDA-approved clinical trial?*

Although many SCI researchers believe that there should be independent replication of findings before going forward with a clinical trial (Kwon et al., 2010), the fact is that this is not a requirement for Food and Drug Administration (FDA) approval of early phase clinical trials. Despite extensive coverage in the media and discussion within the patient and scientific communities, there has been no published independent replication of the original report describing enhanced recovery following transplants of oligodendrocyte precursors derived from human embryonic stem cells (Keirstead et al., 2005). Nevertheless, a trial for Geron's stem cells was approved by the FDA and patients have begun to receive transplants. This example illustrates one of the differences between academics, federal funding agencies, and the FDA, which regulates clinical trials and ultimately approves drugs and other therapeutics. It also highlights an important difference between privately vs. publically funded trials; preclinical data or even the results of completed trials may not be made public or subjected to peer-review by pharmaceutical and biotechnology companies that fund their own trials. Thus replication studies may be carried out but not made public before a clinical trial is allowed to proceed.

Some of the agents tested in the replication projects are also moving towards or are already in clinical trials. For example, minocycline is being tested in a pilot study to evaluate the feasibility of a clinical trial in patients with acute SCI (Casha et al., NCT00559494). Also, clinical studies testing the tolerability and efficacy of erythropoietin have been initiated in patients with spinal shock and spinal cord compression. However, both of these studies have been either terminated or suspended (NCT0020675, NCT00561067). Thus, it will be interesting to follow the results of these clinical investigations.

*Do replicability and robustness in preclinical data predict success in translation?*

Part of the motivation for the FORE—SCI program was to explore whether lack of replication was actually a bottleneck for translating discoveries into preclinical development. One opinion is that the “promising” basic science discoveries are important in terms of advancing knowledge but are not stepping stones to therapy. To paraphrase a comment made by Mark Tuszynski (personal communication): “If there was a really robust treatment out there for spinal cord injury, it would be in clinical trials”. In other words, despite apparent progress on the basic science side and speculations about translation now commonly seen in the Discussion sections of publications, the effects reported so far have not been proven to be robust and translatable. This raises the issue that one hears discussed in the halls: “how many points change in BBB scores is really meaningful”?

There is an increasing recognition of the issue of reproducibility and effect size in preclinical studies. A recent paper in the Policy

Forum of Science Magazine (London et al., 2010) sets out some of the critical questions "that all translational fields must ask".

- Do preclinical studies provide credible measure of effect?
- Are the findings internally valid and replicated independently?
- Is there reasonable external validity?
- Are findings proven to be robust and generalizable?
- Are choices of animal models and outcome measures sufficiently justified?
- Is the assumption that observations in the preclinical system will be reproduced in human patients justified?
- Do data and literature show causal preconditions of effect in animals are also present in humans?
- Do conditions used in preclinical studies correspond to those in the proposed human trial?
- Are delivery strategies, targets, doses, and materials in the human study identical or substantially equivalent to those validated in animals?"

It is our view that treatments must produce robust effects in order to be considered ready for translation. There are infinitely imaginable experimental details that may explain results obtained in replication studies, and methods may never be fully replicated in two different laboratories. Nonetheless, the efforts of the FORE—SCI program to faithfully replicate the published methods reported in the SCI literature have raised important questions that the original authors and the field as a whole should consider. Thus we feel that the replication project has served an important purpose, by identifying reportedly therapeutic approaches that require far more scrutiny and refinement if they are ever to be considered ready for testing in clinical trials.

*Some common myths*

Our combined experience has revealed that some dire expectations expressed at the time the program was initiated were fallacies. For example:

1) It's hard/impossible to publish negative results. This is certainly false. Reviewers have been very positive. Every replication paper has been accepted and this special issue highlights the recognition that such publications are important contributions to the body of scientific knowledge. There are other examples of recent publications where negative results are published in high impact journals (Lee et al., 2010).
2) Repeating an experiment is not interesting science, especially if the results are negative. The FORE—SCI program was initiated with the assumption that most findings would be confirmed and, though confirmatory studies are not "innovative" they would provide information essential to the translational process. The expectation that negative results are not interesting is clearly false. There is increasing recognition that reporting negative results is important and interesting (Sena et al., 2010) and there have been several unexpected methodological and other findings discovered during FORE—SCI replications that have added value.
3) You'll make enemies. It is rewarding for a scientist to learn that their findings have been independently verified, and perhaps distressing to learn the opposite. Nevertheless, our experience has been that the authors of papers selected for replication have exhibited a cooperative spirit with the aim of getting it right. If a given finding is not replicated, that does not necessarily mean that the original study was flawed. In most cases, it seems more likely that the reported effect depends on the exact experimental conditions. This means that translation may be premature or unwarranted, but in many cases, the aim of the original papers was biological discovery, rather than pre-clinical development. In this regard, the replication experiments help define the limits and parameters of the biological phenomena. In spite of the obvious concern about differing results, interactions with authors of the original papers have remained cordial and cooperative.

*What we need to do as a field*

The science being done now and rightfully being celebrated is leading to important new insights into biological processes that are important to our understanding of SCI. At the same time, speculation about therapeutic opportunity has become commonplace. Demonstrating that something is important from the biological standpoint does not necessarily mean that it is a reliable target for therapeutic development or that the effect of a novel therapeutic strategy is robust enough to be described as promising for translation in early publications of the phenomenon. It is troublesome to hear the comment: "we had to work hard to get the experiment to work". This suggests that the effect being demonstrated is fragile and highly dependent on the exact experimental conditions or statistical analyses. The more precise the conditions need to be to demonstrate an effect in a specific animal model, the less likely it is that the phenomenon will represent a reasonable target for therapy.

For those interested in translation, it might be useful to carry out different experiments in which there is a deliberate effort to vary overall study conditions even if this leads to increased variance, and report the results as confirmation or refinement of the experimental target mechanisms. An example of this strategy is a recently funded Department of Defense program where four individual laboratories will test specific drug treatments using different traumatic brain injury models. Various experimental drug treatments will be screened and tested using a spectrum of injury types and severities (Kochanek et al., in press). This approach has long been recommended in other diseases including stroke and TBI (STAIR, 1999; Narayan et al., 2002). In addition, guidelines have been drafted in the field of SCI (Anderson et al., 2005) and other disease areas that include recommendations for preclinical animal studies and reporting (Redmond and Freeman, 2001; Ludolph et al., 2010; Sena et al., 2007), a process already in place for clinical trials under the CONSORT Statement (Schulz et al., 2010).

Confidence in a finding can be increased if there is an independent replication of key findings as part of an original study. For example, Dr. Zhigang He arranged for an independent replication of a key experiment demonstrating enhanced regeneration of CST axons following conditional deletion of PTEN in the sensorimotor cortex (Liu et al., 2010). The independent replication was carried out by a different team of investigators (Dr. Binhai Zheng's team) at a different institution, and the results were then included in the paper.

We need to publish negative results. Although it is sometimes painful to write up studies where results are negative, the findings are important to the field. Science is an iterative process wherein questions raised about previous theories and studies must be addressed to bring greater understanding and confidence to the conclusions being drawn.

As reviewers and editors, we need to be more insistent about providing full methodological details. For studies of SCI, it is important to explain more fully how groups were compiled and over what time frame, the degree to which timing of surgery and functional testing was randomized, how blinding was accomplished, etc. This is not just to make replication easier. The main purpose is to allow a more refined judgment by readers about the possibility of unintended bias.

## Acknowledgments

Supported by N01-NS-3-2353 and HHSN271200800039C (O.S.), HHSN271200800040C (P.G.P.), and N01-NS-3-2352 (W.D.D.). The authors gratefully acknowledge Ms. Laurie Leonard, who serves as Contracting Officer for the FORE—SCI contracts and Dr. Arlene Chiu, who initiated the NINDS FORE—SCI program.

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